Examination –June- 2023
B.Tech. I/II Sem : CSE, IT, AI, BC / EE, EI, EC, IoT
Applied Chemistry

Time :2 Hrs

Max. Marks : 60
Min. Marks :19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c &d has internal Choice. Assume missing data, if any.
Word limit be observed as follows:
Part a & b – Objective type/Fill in the blanks/One sentence answer.
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

Q.1 (a) Following two questions have four options, out of which only one is correct. Write the correct answer only. 02

(i) A water sample containing hardness in range of 0 to 60 ppm, may be categorized as
(1) Hard water (2) Very Hard water (3) Soft Water (4) Moderately hard water

(ii) Which of the following is NOT a softening method
(1) Sterlisation using Bleaching Powder (2) Lime Soda method
(3) Zeolite method (4) Ion Exchange Resin method

(b) 500 ml of water sample contains 0.292 g of calcium oxalate monohydrate. Express calcium concentration in ppm. 02

(c) Why is hardness of water expressed in terms of $CaCO_3$ equivalent? Give relationship between different units for expressing hardness of water. 03

OR

Write the formula and names of substances which impart temporary and permanent hardness to water. 03

(d) Explain Ion Exchange Resin method of water softening giving only: 05

(i) Well labelled diagram
(ii) Chemical reactions involved including regeneration of resins
(iii) Hardness of soft water obtained in ppm

OR

A hard water sample has following compounds: 0

$Ca(HCO_3)_2$= 162 ppm; $Mg(HCO_3)_2$= 73 ppm; $CaCl_2$= 111 ppm; $MgCl_2$= 95 ppm; $CaSO_4$= 136 ppm; $MgSO_4$= 120 ppm; CO_2= 44 ppm.

Calculate Temporary and Permanent Hardness of water. Also calculate the quantity of lime and soda required for softening 100000 litres of this water.

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Q.2 (a) Following two questions have four options, out of which only one is correct. Write the correct answer only.

(i) A galvanic cell converts

(1) Electrical energy into chemical energy (2) Chemical energy into heat energy

(3) Chemical energy into electrical energy (4) Electrical energy into heat energy

(ii) Which is correct regarding efficiency of various batteries

(1) Lead battery > Galvanic cell > Li ion (2) Galvanic cell > Li ion > Lead battery

((3) Li ion > Galvanic cell > Lead battery (4) Li ion > Lead battery >Galvanic cell

(b) Write the reactions taking place in Lead accumulator battery during charging and discharging. 02

(c) Write construction and working of Normal Hydrogen electrode 03

OR

An electrochemical cell is represented as 03

$Zn_{(s)} | Zn^{2+}_{(0.1M)} || Cu^{2+}_{(0.1M)} | Cu_{(s)}$

Calculate the emf of the above cell at 25°C if $E^{o}_{Zn2+/Zn} = -0.75$ V and $E^{o}_{Cu2+/Cu} = +0.34$ V

(d) Draw a well labeled diagram of Lithium ion batteries. Also write the reactions taking place at anode and cathode. Compare the efficiency of Lithium ion batteries with other batteries. 05

OR

Derive Nernst's equation for electrode potential of a cell. Write importance and applications of Nernst's equation. 05

Q.3 (a) Following two questions have four options, out of which only one is correct. Write the correct answer only. 02

(i) When a buried pipeline is protected from corrosion by connecting to Mg block, it is called

1. impressed voltage protection 2. sacrificial anodic corrosion
3. sacrificial cathodic corrosion 4. passivity

(ii) Electrochemical corrosion can only occur only if

1. Oxygen is present in contact with metal
2. Air is present in contact with metal
3. Liquid medium is present in contact with metal
4. Nitrogen is present in contact with metal

(b) Why does corrosion of water filled iron tanks occur below the waterline? 02

(c) Differentiate between corrosion and protection of metals. 03

OR

Explain briefly various factors affecting corrosion. 03

(d) Discuss various methods applied for protection of metals from corrosion. 05

OR

Explain the mechanism of Electrochemical corrosion. How does it differ from Chemical corrosion? 05

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Q.4 (a) Following two questions have four options, out of which only one is correct. Write the correct answer only. 02

(i) Polymerisation in which two or more chemically different monomers take part, is called

1. Addition Polymerisation 2. Condensation Polymerisation
3. Co Polymerisation 4. Chain Polymerisation

(ii) For making rechargeable batteries, which polymeric material is generally used?

1. Polypropylene 2. Polyglycol 3. Polypeptide 4. Polyaniline

(b) What are photoactive polymers? Give its uses. 02

(c) What are electroactive polymers? Give examples. Write their applications. 03

OR

What are nanomaterials? Give examples. Write important applications of nanomaterials in modern technologies. 03

(d) Write method of preparation and properties of *any two* of the following: 0

1. Polypyrrol 2. Polyaniline 3. Polythiophene

OR

Write brief notes on *any two* of the following: (

1. Optical Fibers 2. Liquid Crystal Polymers 3. Graphene

Q.5 (a) Following two questions have four options, out of which only one is correct. Write the correct answer only. (

(i) Reciprocal of wavelength of a radiation is called

1. Frequency 2. Wave Number 3. Density 4. Velocity

(ii) Which of the following expresses correct pH for entire range of pH scale

1. <7 Basic; 0 Neutral: >7 Acidic 2. >7 Acidic; 7 Neutral; <7 Basic
3. >7 Basic; 7 Neutral; <7 Acidic 4. <7 Acidic; 0 Neutral; >7 Basic

(b) Selection of an appropriate filter (having a specific wavelength) is required in colorimetry. Why?

(c) Draw a well labelled diagram of instrument used in Gas Chromatography.

OR

Draw a well labelled diagram of the instrumental set up used in conductivity measurement.

(d) State and derive Beer Lambert's law. How Beer Lambert's law can be verified colorimetrically using $KMnO_4$ solution.

OR

Define pH value. What is pH scale? How pH of a solution is determined using pH meter?

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